

The Replica Method I: A Single-Spin Toy Model

Physics of Learning • IIT Madras

Chapter 4

We introduce the replica method—the central technique for computing disorder-averaged free energies in disordered systems and in the statistical mechanics of learning. The whole conceptual apparatus is developed on the simplest nontrivial example: a single binary neuron in a random field. Every intermediate step is shown. By the end we have computed $\ln Z$ two independent ways (validating the method) and learned to *read* the free energy as a statement about how strongly data determines a neuron. This chapter is the conceptual scaffold for Chapter 5, where the same steps—applied to a system of N coupled spins—produce genuine phase transitions in learning.

Why we must average a logarithm

IN A LEARNING PROBLEM the data, or the couplings built from it, are random but *fixed* for a given task.¹ For one such fixed sample, the thermodynamics is governed by the free energy

$$F = -\frac{1}{\beta} \ln Z, \quad Z = \sum_{\{\mathbf{s}\}} e^{-\beta H[\mathbf{s}]}, \quad \beta = \frac{1}{T}.$$

Physical observables are *self-averaging*: a large system behaves like the typical sample, i.e. like the average over disorder. The object we actually need is therefore the disorder-averaged free energy

$$\bar{F} = -\frac{1}{\beta} \overline{\ln Z},$$

where $\overline{(\cdot)}$ denotes the average over the quenched randomness.

HERE IS THE DIFFICULTY. Averaging Z itself over disorder is typically easy—it is a Gaussian integral. But averaging $\ln Z$ is hard: the logarithm entangles the disorder with the thermal sum in a way that refuses to factorise. The replica method is the standard device for circumventing exactly this obstruction.

The replica identity

The escape hatch is an elementary algebraic identity:

¹ This is called *quenched* disorder, as opposed to *annealed* disorder which fluctuates on the same timescale as the degrees of freedom. The distinction is the whole reason the replica method is needed.

$$\ln Z = \lim_{n \rightarrow 0} \frac{Z^n - 1}{n}.$$

Verification. Expand $Z^n = e^{n \ln Z} = 1 + n \ln Z + O(n^2)$. Then $(Z^n - 1)/n = \ln Z + O(n) \rightarrow \ln Z$ as $n \rightarrow 0$. $\square$

Averaging both sides converts the hard object into

$$\overline{\ln Z} = \lim_{n \rightarrow 0} \frac{\overline{Z^n} - 1}{n}.$$

The point is that $\overline{Z^n}$ —the average of the n -th *power*—is computable. For integer n , Z^n is simply n identical, independent copies (“replicas”) of the system:

$$Z^n = \underbrace{Z \cdot Z \cdots Z}_n = \sum_{\{s^1\}} \cdots \sum_{\{s^n\}} e^{-\beta \sum_{a=1}^n H[s^a]}.$$

The replicas $s^1, \dots, s^n$ are identical copies that *share the same disorder*. When we average over that shared randomness, it couples the replicas together—and that coupling is the entire mechanism.

THE SLEIGHT OF HAND is this: we compute $\overline{Z^n}$ for *integer* n , obtain a formula, then *analytically continue* it to real n and send $n \rightarrow 0$.²

The toy model: one neuron in a random field

Take a *single spin* $s = \pm 1$ in a random field h , where h is Gaussian with zero mean and variance σ^2 :

$$H[s] = -hs, \quad s = \pm 1, \quad h \sim \mathcal{N}(0, \sigma^2).$$

Interpret $s = +1$ ($\uparrow$) as the neuron *firing* and $s = -1$ ($\downarrow$) as *silent*. The field h is a random input or bias—it plays the role of the quenched disorder, “the data.” The energy $-hs$ is lowest when s aligns with $\text{sign}(h)$: the neuron tries to match its input.

For a fixed field h the partition function is immediate:

$$Z = \sum_{s=\pm 1} e^{\beta hs} = e^{\beta h} + e^{-\beta h} = 2 \cosh(\beta h).$$

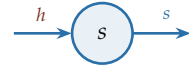
Our goal is the disorder average $\overline{\ln Z}$, with the average taken over h .

The direct route (our ground truth)

Because this model is so simple, we can average the logarithm directly:

$$\overline{\ln Z} = \int_{-\infty}^{\infty} \frac{dh}{\sqrt{2\pi\sigma^2}} e^{-h^2/2\sigma^2} \ln(2 \cosh \beta h). \quad (1)$$

² This continuation is the mathematically delicate step. In this chapter it is completely harmless; in Chapter 5 it becomes the subtle heart of the method.



A single neuron $s = \pm 1$ driven by a random field h .

This is a perfectly good one-dimensional integral. We keep Eq. (1) in reserve: at the end we will reproduce it via replicas, confirming the method on a case where we already know the answer.

The replica route, step by step

We now pretend we do *not* know how to average the log, and proceed through the replica machinery.

Step 1 — Replicate

For integer n , write n copies all feeling the *same* field h :

$$Z^n = \prod_{a=1}^n \sum_{s^a=\pm 1} e^{\beta h s^a} = \sum_{\{s^a\}} \exp\left(\beta h \sum_{a=1}^n s^a\right). \quad (2)$$

The replica index $a = 1, \dots, n$ is the new bookkeeping device. Every replica sees the same h ; that shared disorder is what will couple them.

Step 2 — Average over disorder

The disorder h appears *linearly* in the exponent of Eq. (2), so the average is a Gaussian integral. Using the identity³ $\overline{e^{hx}} = e^{\sigma^2 x^2/2}$ with $x = \beta \sum_a s^a$,

$$\overline{Z^n} = \sum_{\{s^a\}} \exp\left(\frac{\sigma^2 \beta^2}{2} \left(\sum_{a=1}^n s^a\right)^2\right). \quad (3)$$

The field has vanished, but it has left behind a *square* of the spin sum. That square is the seed of all replica coupling.

Step 3 — The coupling between replicas

Expand the square, using $(s^a)^2 = 1$:

$$\left(\sum_a s^a\right)^2 = \sum_a (s^a)^2 + \sum_{a \neq b} s^a s^b = n + \sum_{a \neq b} s^a s^b.$$

Hence

$$\overline{Z^n} = e^{\sigma^2 \beta^2 n/2} \sum_{\{s^a\}} \exp\left(\frac{\sigma^2 \beta^2}{2} \sum_{a \neq b} s^a s^b\right).$$

The term $s^a s^b$ is an *effective interaction* between replicas a and b —even though the original spins never interacted.⁴ Disorder averaging has manufactured an interaction out of nothing.

³ $\overline{e^{hx}} = \int \frac{dh}{\sqrt{2\pi\sigma^2}} e^{-h^2/2\sigma^2} e^{hx} = e^{\sigma^2 x^2/2}$, the moment generating function of a Gaussian.

⁴ In the many-spin systems of Chapter 5 this object becomes the *overlap order parameter* $q_{ab} = \frac{1}{N} \sum_i s_i^a s_i^b$, the central quantity of the entire theory. Here, with a single spin, it is merely the number $s^a s^b$.

Step 4 — Decouple by Hubbard–Stratonovich

The summand in Eq. (3) depends on the spins only through $m \equiv \sum_a s^a$. Linearise the square with an auxiliary Gaussian variable:⁵

$$e^{\frac{\sigma^2 \beta^2}{2} m^2} = \int_{-\infty}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} e^{\sigma \beta z m}.$$

⁵ The Hubbard–Stratonovich identity $e^{\frac{1}{2} a m^2} = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} e^{\sqrt{a} z m}$ trades a square in m for a linear term, at the cost of one extra integral. It recurs throughout the subject.

Substituting and pushing the spin sum inside, the replicas factorise:

$$\overline{Z}^n = \int \mathcal{D}z \prod_{a=1}^n \left(\sum_{s^a=\pm 1} e^{\sigma \beta z s^a} \right), \quad \mathcal{D}z \equiv \frac{dz}{\sqrt{2\pi}} e^{-z^2/2},$$

and each replica contributes the same factor $\sum_{s=\pm 1} e^{\sigma \beta z s} = 2 \cosh(\sigma \beta z)$. Therefore

$$\overline{Z}^n = \int_{-\infty}^{\infty} \mathcal{D}z (2 \cosh \sigma \beta z)^n.$$

This is exact for every integer n . The n replicas, decoupled by the field h , have been re-coupled through the single shared variable z (indeed $z = h/\sigma$ here).

Step 5 — Continue, and take $n \rightarrow 0$

We need $\overline{\ln Z} = \lim_{n \rightarrow 0} (\overline{Z}^n - 1)/n$. Observe that $\overline{Z}^n|_{n=0} = \int \mathcal{D}z (\cdot)^0 = \int \mathcal{D}z = 1$, so the limit is simply a derivative at $n = 0$:

$$\overline{\ln Z} = \frac{\partial}{\partial n} \overline{Z}^n \Big|_{n=0}.$$

With $g(z) = 2 \cosh \sigma \beta z$ we have $g^n = e^{n \ln g}$ and $\partial_n g^n = g^n \ln g$, which at $n = 0$ gives $g^0 \ln g = \ln g$. Hence

$$\overline{\ln Z} = \int_{-\infty}^{\infty} \mathcal{D}z \ln(2 \cosh \sigma \beta z).$$

Step 6 — Check against the direct route

Substitute $h = \sigma z$ (so $dh = \sigma dz$ and $h^2/2\sigma^2 = z^2/2$) in the direct result (1):

$$\overline{\ln Z}_{\text{direct}} = \int \frac{dh}{\sqrt{2\pi\sigma^2}} e^{-h^2/2\sigma^2} \ln(2 \cosh \beta h) = \int \mathcal{D}z \ln(2 \cosh \sigma \beta z).$$

Identical to the replica result. The replica method, with its dubious-looking $n \rightarrow 0$ continuation, has produced the correct answer.

Reading the free energy

Expand for small β to extract the physics. Using $\ln(2 \cosh x) \approx \ln 2 + \frac{x^2}{2}$ and $\overline{z^2} = 1$:

$$\overline{\ln Z} \approx \ln 2 + \frac{\sigma^2 \beta^2}{2}. \quad (4)$$

Each term has a clear meaning.

THE $\ln 2$ TERM is the entropy of an *undecided* neuron. At $\beta \rightarrow 0$ or $\sigma \rightarrow 0$ (no signal), $\overline{\ln Z} \rightarrow \ln 2 = \ln(\text{number of states})$ for one binary neuron. With no data signal, the neuron carries exactly one bit of undetermined freedom: on and off are equally likely.

THE $\sigma^2 \beta^2 / 2$ TERM measures how strongly the data pins the neuron down. To see it as an observable, compute the neuron's mean alignment with its field:⁶

$$\langle s \rangle = \tanh(\beta h), \quad \overline{\langle s \rangle^2} = \overline{\tanh^2(\beta h)} \approx \sigma^2 \beta^2 \quad (\text{small } \beta),$$

which is exactly twice the correction term. It quantifies how reliably the neuron's output is determined by its input: near zero when $\sigma\beta$ is small (output essentially random), saturating to 1 as $\sigma\beta \rightarrow \infty$ (output fully fixed by the data).

THE CROSSOVER between these regimes is the toy precursor of a learning transition. As $\sigma\beta$ increases,

$$\sigma\beta \rightarrow 0 : \overline{\ln Z} \rightarrow \ln 2 \text{ (1 bit free)}, \quad \sigma\beta \rightarrow \infty : \overline{\ln Z} \rightarrow \sigma\beta \sqrt{\frac{2}{\pi}} \text{ (locked to data)},$$

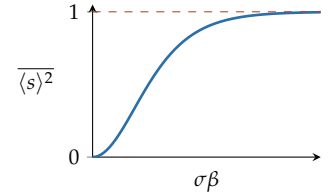
using $\ln(2 \cosh x) \rightarrow |x|$ and $\overline{|z|} = \sqrt{2/\pi}$. Here the crossover is perfectly smooth. In a network of many coupled spins it will *sharpen into a genuine phase transition*—the subject of Chapter 5.

What this chapter established

The logical skeleton, which survives intact into real spin glasses and machine-learning calculations, is:

1. We want $\overline{\ln Z}$; it is intractable because the logarithm does not factorise over disorder.
2. Replace it via $\ln Z = \lim_{n \rightarrow 0} (Z^n - 1)/n$, so we instead need $\overline{Z^n}$.
3. For integer n , Z^n is n replicas sharing the disorder; averaging is a tractable Gaussian integral.
4. Disorder averaging *couples* the replicas, producing $s^a s^b$ —whose many-spin analogue is the overlap q_{ab} .

$$^6 \langle s \rangle = \frac{\sum_s s e^{\beta h s}}{\sum_s e^{\beta h s}} = \frac{e^{\beta h} - e^{-\beta h}}{e^{\beta h} + e^{-\beta h}} = \tanh(\beta h).$$



The single-neuron learning crossover: output determinacy rises from 0 (undetermined) toward 1 (data-locked) as the signal $\sigma\beta$ grows.

5. Evaluate $\overline{Z}^n$ as a function of n , continue to real n , and take $n \rightarrow 0$.

WHAT MADE THIS CASE EASY was the absence of two ingredients. There was no thermodynamic limit ($N \rightarrow \infty$), so no saddle point: the disorder average was an exact finite integral. And the overlap $s^a s^b$ was a single number, not a matrix we had to solve for. Restore both—many coupled spins, and a matrix-valued overlap q_{ab} determined self-consistently—and the replica method begins to do work that nothing else can: it predicts the storage capacity of a memory, the shape of a learning curve, and the sharp transitions between learnable and unlearnable regimes. That is Chapter 6.